Data Communication & Computer Networks (CN)

Topic 3 of 7 : Addressing (MAC, IPv4, IPv6, Port, DNS)

Curation in progress — 77 auto-extracted questions pending verification (see Pending Review at the end)

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • RSMSSB IA • RSMSSB SA • Data Entry/DD • BCI • SCI • REET • RAS & other Rajasthan Govt. Exams**

Total Questions: **0** verified MCQs + **77** pending review

Question Type: MCQ — every question has 4 options (A, B, C, D)

Languages: **Bilingual:** English + Hindi (हिन्दी)

Every question includes: ✔ Correct answer (why it is RIGHT) • ★ Nukta / key point (exam tip) • ✖ why each wrong option is WRONG

Verification: Every answer cross-verified **4 times** (Forouzan facts + official answer keys of Subject Test / Unit Test Solutions where available)

Syllabus Coverage — sub-topic wise order

Sub-topic curation in progress — auto-extracted questions are listed in the **"Pending Review"** section at the end.

How to read this PDF: Each question shows 4 options (A-D) in English and Hindi. The line starting with ✔ gives the correct answer and why it is right. The line starting with ★ is the **Nukta** — the key point to remember for the exam. The lines starting with ✖ explain why every wrong option is wrong. Read the question, try it yourself, then check the answer.

System: This PDF is generated by **tools/cn_build_topic.py** from **tools/data/topic3_data.py**. To add more questions, drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions are auto-classified into topics and appended here (as pending) for review. See **Computer Network/README.md**.

Pending Review — auto-ingested, needs verification

These questions were auto-extracted & classified from uploaded PDFs. Answers/nukte to be verified before counting.

⚠ NEW — auto-ingested, answer needs verification (pending review)

P1. Which of the following port numbers belong to the “well-known” ports category? "well-known"

(A) 80, 443, 8080
(B) 49152, 110, 53
(C) 22, 25, 80
(D) 1024, 3306, 8080

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P2. IP address 127.0.0.1 refers to which function? IP 127.0.0.1 function

(A) Broadcast
(B) Class D network
(C) Loopback testing
(D) APIPA

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P3. A host has IP 192.168.10.45 /27. Which statement is TRUE? Host IP 192.168.10.45

(A) Broadcast address = 192.168.10.63
(B) Network ID = 192.168.10.32
(C) First usable host = 192.168.10.33
(D) All of the above

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 3 / MAC Address]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P4. A subnet /30 is MOST commonly used for: Subnet

(A) Two host networks
(B) Four host networks
(C) Loopback only
(D) VPN networks

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P5. Which of the following is a class D address? - Class D address

(A) 224.1.1.1
(B) 233.0.0.10
(C) 239.5.5.5
(D) All of the above

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P6. Which range is typically used for ephemeral (dynamic) ports? ephemeral (dynamic) ports

(A) 0–1023
(B) 1024–49151
(C) 49152–65535
(D) 100–200

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P7. Which of the following port and IP address scanner is famous among users? # % B - 9 IP S „ 9 - †

(A) Cain and Abel
(B) Angry IP Scanner
(C) Snort
(D) Ettercap

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P8. Which of the following uniquely identifies a process on a host? # B “ ”

(A) IP Address only
(B) Port Number only
(C) IP Address + Port Number
(D) MAC Address

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P9. Transport layer uses which addressing? E A - E

(A) IP address
(B) MAC address
(C) Port number
(D) Domain name

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P10. DNS works on which layer? DNS

- **(A)** Transport
- **(B)** Application
- **(C)** Network
- **(D)** Data link

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P11. What is the usual length of a MAC address? MAC E k R

- **(A)** 16 bits
- **(B)** 32 bits
- **(C)** 48 bits
- **(D)** 64 bits

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / MAC Address]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P12. The first octet of a Class A IP address ranges from: Class A IP E W m 7 R

- **(A)** 0-127
- **(B)** 128-191
- **(C)** 192-223
- **(D)** 224-239

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P13. How many bits are used for the network portion in a Class C IP address? Class C IP E 7 A O

- **(A)** 8
- **(B)** 16
- **(C)** 24
- **(D)** 32

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P14. Which class of IP address is used for multicasting? f :n o IP E

- **(A)** Class A
- **(B)** Class B
- **(C)** Class C
- **(D)** Class D

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P15. Which class of address is reserved for experimental purposes? IP E o h T 7 p q r

(A) Class C
(B) Class B
(C) Class D
(D) Class E

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P16. Which address identifies a unique device on a local network segment? s A 7 7 E

(A) IP address
(B) MAC address
(C) Gateway
(D) Subnet Mask

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P17. Which is a static hardware address? :s A E ?**

(A) IP address
(B) MAC address
(C) Port number
(D) Domain name

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / MAC Address]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P18. MAC address is also known as: MAC E ?**

(A) Logical address
(B) Hardware address
(C) IPv4 address
(D) Routing address

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / MAC Address]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P19. IP address is divided into how many parts? IP E 7 ?**

(A) 2
(B) 3
(C) 4
(D) 5

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P20. What is the default subnet mask for Class B network?Class B

(A) 255.0.0.0
(B) 255.255.0.0
(C) 255.255.255.0
(D) 255.255.255.255

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P21. Which IP class is reserved for multicasting? IP

(A) Class A
(B) Class B
(C) Class D
(D) Class E

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P22. In IPv6, how many bits are in the address?IPv6

(A) 32
(B) 64
(C) 128
(D) 256

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / IPv6 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P23. Which of these DNS record types maps a domain name to an IP address? - DNS IP

(A) CNAME
(B) A
(C) MX
(D) PTR

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P24. Which part of IP address defines the network in Class C?Class C IP

(A) First octet
(B) First two octets
(C) First three octets
(D) All four octets

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P25. What is the default port number for DNS?DNS

(A) 80
(B) 21
(C) 53
(D) 110

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P26. In IPv6, which of the following is true?IPv6

(A) Uses 32-bit address
(B) Includes checksum in header
(C) Supports anycast addressing
(D) Single amplification

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / IPv6 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P27. Which address type is unique in the network and globally routable in IPv6?IPv6

(A) Link-Local
(B) Multicast
(C) Unique local
(D) Global unicast

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / IPv6 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P28. What is the maximum number of hosts in a /24 subnet? 24

(A) 128
(B) 256
(C) 254
(D) 512

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P29. What is the typical range of ephemeral ports?Ephemeral

(A) 0-1023
(B) 1024-49151
(C) 49152-65535
(D) None of the these

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P30. subnet mask of 255.255.255.248 allows how many usable hosts?255.255.255.248 usable

- **(A)** 2
- **(B)** 6
- **(C)** 14
- **(D)** 30

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P31. What is the function of IGMP in networking? IGMP

- **(A)** Manages IP address allocation
- **(B)** Routes datagrams
- **(C)** Manages multicast group membership
- **(D)** Handles device discovery

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / MAC Address]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P32. What is the maximum number of IP addresses in Class C?Class C IP

- **(A)** 254
- **(B)** 256
- **(C)** 1024
- **(D)** 65536

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P33. Which type of DNS record maps IP to hostname? - DNS IP

- **(A)** MX
- **(B)** A
- **(C)** PTR
- **(D)** CNAME

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P34. In subnetting, reducing the number of host bits does what

- **(A)** Increase broadcast domain
- **(B)** Decrease number of subnets
- **(C)** Increase number of subnets
- **(D)** all Increase number of hosts

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P35. What type of address is FF:FF:FF:FF:FF:FF?FF:FF:FF:FF:F F:FF

- **(A)** Unicast
- **(B)** Multicast
- **(C)** Boardcast
- **(D)** Loopback

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / MAC Address]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P36. MAC address is also called _____. _____

- **(A)** Physical address
- **(B)** Logical address
- **(C)** Source address
- **(D)** Destination address

[Source: CN Topic Test-01.pdf | auto-classified: Topic 3 / MAC Address]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P37. What is the main purpose of a Domain Name System (DNS) server? (DNS)

- **(A)** To allocate IP addresses dynamically
- **(B)** To convert domain names into IP addresses
- **(C)** To route traffic within a LAN
- **(D)** To manage firewall settings

[Source: CN Topic Test-02.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P38. There is a need to create a network that has 5 subnets, each with at least 16 hosts. which one is used as classful subnet mask? 16

- **(A)** 255.255.255.192
- **(B)** 255.255.255.248
- **(C)** 255.255.255.240
- **(D)** 255.255.255.224

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P39. Which of the following is a class B host address

- **(A)** 230.0.0.0
- **(B)** 130.4.5.6
- **(C)** 230.7.6.5
- **(D)** 30.4.5.6

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P40. A subnet mask in class C can have _____ 1's with the remaining bits O'sb C _____ 1

(A) 10
(B) 24
(C) 12
(D) 7

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P41. IPv6 is developed by IPv6

(A) IETF
(B) ANSI
(C) ISO
(D) IEEE

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv6 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P42. In the IPv4 addressing format, the number of networks allowed under Class C addresses is IPv4 , C

(A) 220
(B) 224
(C) 214
(D) 221

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P43. Which of the following statement/s is/are true? (i) Windows XP supports both peer- peer and client-server networks. (ii) Windows XP implements Transport protocols as drivers that can be loaded and uploaded from the system dynamically. (i) (ii) Choose the correct answer from the code given below:

(A) Both (i) and (ii)
(B) Neither (i) nor (ii) (ii) only (i) only (ii) 11. The four byte IP Address consists of (a) Neither network nor Host Address (b) Network Address
(C) Both Network and Host Address
(D) Host Address

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P44. The dotted-decimal notation of the following IPV4 address in binary notation is IPV4

(A) 111.56.45.239
(B) 129.11.10.238
(C) 129.11.11.239
(D) 111.56.11.239

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P45. What is the maximum number of hosts that can be assigned IP addresses on a /24 network? 24 IP

(A) 254
(B) 256
(C) 512
(D) 1024

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P46. In a class B subnet, we know the IP address of one host and the mask as given below: IP address: 125.134.112.66 Mask: 255.255.224.0 What is the first address (Network address)

(A) 125.134.96.0
(B) 125.134.112.0
(C) 125.134.112.66
(D) 125.134.0.0

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P47. In CIDR notation, what does the /16 represent? CIDR

(A) 16 host bits
(B) 16 network bits
(C) 16 bytes
(D) 16 subnets

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P48. Which IP address class provides the largest number of hosts per network

(A) Class A
(B) Class B
(C) Class C
(D) Class D

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P49. If the number of networks and number of hosts in class B are 2m, (2n −2) respectively. Then the relation between m,n is: 2m (2n −2) m,n :

(A) 3m=2n
(B) 7m=8n
(C) 8m=7n
(D) 2m=3n

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P50. The address 255.255.255.255 is used for:

(A) A public IP
(B) A default gateway
(C) A network ID
(D) Broadcast to all nodes

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P51. What is the range of addresses for APIPA (Automatic Private IP Addressing)? APIPA ( )

(A) 169.254.0.1 to 169.254.255.254
(B) 10.0.0.0 to 10.255.255.255
(C) 192.168.0.0 to 192.168.255.255
(D) 172.16.0.0 to 172.31.255.255

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P52. Which part of the IP address 172.16.5.4/24 identifies the network? 172.16.5.4/24

(A) 172
(B) 172.16
(C) 172.16.5
(D) 172.16.5.4

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P53. The first octet of a Class B IP address falls between:

(A) 0-127
(B) 128-191
(C) 192-223
(D) 224-239

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P54. What is the purpose of subnetting

(A) To increase the number of hosts per network
(B) To divide a network into smaller networks
(C) To reduce the number of IP addresses needed
(D) To merge multiple networks

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P55. What is the valid host range for the subnet 192.168.1.0/25? 192.168.1.0/25

(A) 192.168.1.0 - 192.168.1.255
(B) 192.168.1.1 - 192.168.1.126
(C) 192.168.1.1 - 192.168.1.254
(D) 192.168.1.0 - 192.168.1.127

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P56. In a subnet mask, the number of 1's represents:

(A) Host bits
(B) Network bits
(C) Broadcast bits
(D) Subnet bits The IP address 224.0.0.1 belongs to: IP address 224.0.0.1 : (a) Private IP address (b) Public IP address (c) Multicast address (d) Broadcast address

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P57. How many usable hosts are there in a /30 subnet? 30

(A) 2
(B) 4
(C) 6
(D) 14 40 What does NAT (Network Address Translation) primarily help with? NAT ( ) (a) Speeding up IP packets (b) Connecting two networks (c) IP address conservation (d) Encrypting IP traffic

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P58. What is the purpose of the "reserved" IP address 0.0.0.0? reserved IP address 0.0.0.0

(A) Default gateway
(B) Used for DHCP servers
(C) Default route or "any network"
(D) Multicast address

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P59. Which header field is removed when an IP packet moves across a router? IP

- **(A)** Source IP address
- **(B)** Destination IP address
- **(C)** MAC (Media Access Control) address
- **(D)** Subnet mask

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P60. The subnet mask 255.255.255.128 provides how many subnets for a Class C network? 255.255.255.128

- **(A)** 1
- **(B)** 2
- **(C)** 4
- **(D)** 8

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P61. Which IP address type is used to communicate with all devices on a subnet

- **(A)** Unicast
- **(B)** Broadcast
- **(C)** Multicast
- **(D)** Anycast

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / MAC Address]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P62. The address of a class B host o be split into subnets with a 3 bit subnet number. What is the maximum number of subnets and maximum number of hosts in each subnet

- **(A)** 8 subnets and 262141 hosts
- **(B)** 6 subnets and 262141 hosts
- **(C)** 6 subnets and 1022 hosts
- **(D)** None of the Above

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P63. The special IP address used to test network interface (NIC) hardware is: ( )

(A) 127.0.0.1
(B) 0.0.0.0
(C) 255.255.255.255
(D) 169.254.0.1 The process of hiding internal IP addresses from external networks is called: IP (a) Port forwarding (b) NAT (c) Tunneling (d) VPN

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P64. Which class of IP addresses supports 65,534 hosts per network? 65,534

(A) Class A
(B) Class B
(C) Class C
(D) Class D

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P65. Which IP address is reserved for future use and experimental purposes

(A) 127.0.0.1
(B) 224.0.0.1
(C) 240.0.0.0
(D) 192.168.0.1

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P66. An ethernet destination address 0A-01-12-03-04-05 is: 0A-01-12-03-04 05 :

(A) Unicast address
(B) Multicast address
(C) Broadcast address
(D) All of the options

[Source: CN Topic Test-04.pdf | auto-classified: Topic 3 / MAC Address]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P67. The IP address is used by hosts when they are being booted

(A) 0.0.0.0
(B) 1.0.0.0
(C) 1.1.1.1
(D) 255.255.255.255

[Source: CN Topic Test-04.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P68. The address used in the Data Link Layer is called:

(A) Logical address
(B) Port address
(C) MAC address
(D) IP address

[Source: CN Topic Test-04.pdf | auto-classified: Topic 3 / MAC Address]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P69. What does /24 denote in CIDR?

(A) 24 bits are used for host portion
(B) 24 bits are used for network portion
(C) 24 subnets are created
(D) 24 addresses are available in the subnet

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P70. IP address 172.16.123.213 is found to be residing on a sub-network with the following subnet mask: 255.255.255.240. What is the network address of this subnet?

(A) 172.16.123.240
(B) 172.16.123.208
(C) 172.16.123.213
(D) 172.16.123.245

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P71. The size of the 'total length field' of IPv4 datagram is -

(A) 16 bits
(B) 32 bits
(C) 8 bits
(D) 4 bits

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P72. What is the role of DNS? It is best described as -

(A) maps a domain name to an IP address
(B) translate a host's private IP address to a public IP address
(C) translate a host's IPv6 address to an IPv4 address
(D) maps a MAC address to an IP address

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P73. A Reverse Lookup in DNS means a lookup to ..... using ..... records.

(A) identify mail servers, SOA
(B) map IP address to domain name, PTR
(C) map domain names to IP address, A
(D) determine the authoritative name servers, CNAME

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P74. An organization has a Class B network and wishes to form subnets for 64 departments. The subnet mask would be -

(A) 255.255.128.0
(B) 255.255.0.0
(C) 255.255.64.0
(D) 255.255.252.0

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P75. If a Class B network on the internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

(A) 2046
(B) 1022
(C) 1023
(D) 1024

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P76. The Address Resolution Protocol (ARP) is used for -

(A) finding the IP address that corresponds to a MAC address
(B) finding the MAC address that corresponds to an IP address
(C) finding the IP address from the DNS
(D) finding the IP address of the default gateway

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P77. The size of MAC address (in bits) is -

(A) 42
(B) 48
(C) 24
(D) 36

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / MAC Address]